

KENDRIYA VIDYALAYA SANGATHAN ERNAKULAM REGION

ARTIFICIAL INTELLIGENCE (SUBJECT CODE - 417) - CLASS X

PRE BOARD EXAMINATION 2024-25

TIME: 2 Hours

Max.Marks: 50

General Instructions

1. Please read the instructions carefully.
2. This Question Paper consists of 21 questions in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. Out of the given $(5 + 16) = 21$ questions, a candidate has to answer $(5 + 10) = 15$ questions in the allotted (maximum) time of 2 hours.
5. All questions of a particular section must be attempted in the correct order.

SECTION A – OBJECTIVE TYPE QUESTIONS (24 MARKS):

- This section has 05 questions.
- Marks allotted are mentioned against each question/part.
- There is no negative marking.
- Do as per the instructions given.

SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):

- This section has 16 questions.
- A candidate has to do 10 questions.
- Do as per the instructions given.
- Marks allotted are mentioned against each question/part.

SECTION A: OBJECTIVE TYPE QUESTIONS

Q.1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	
i	Sonia gets up at 6 am and goes for her hobby classes. Then she comes back home and finishes her homework before going to school. She does all work by herself. No one tells her to do so. This is called _____. a. Self-Awareness b. Self-Motivation c. Self-Regulation d. Discipline	1
ii	_____ act as an interface between the user and the computer. a. Operating System b. MS Excel c. Impress d. Digital Documentation	1

iii	Assertion (A): An entrepreneur takes risks. Reason (R): An entrepreneur is self-confident. (a) Both A and R are correct and R is the correct explanation of A (b) Both A and R are correct but R is NOT the correct explanation of A (c) A is correct but R is not correct (d) A is not correct but R is correct	1
iv	Which of the below is not a step of Effective Time Management? a. Organize b. Prioritize c. Control d. Remember	1
v	Reenu travelled to Spain from India to pursue her higher education. But she doesn't know how to speak Spanish (language of Spain). Because of this, she was unable to find a part time job. This is an example of _____. a. Interpersonal barrier b. Physical barrier c. Organizational barrier d. Linguistic barrier	1
vi	Solutions to these global problems like scarcity of clean water to drink, scarcity of pure air to breathe, scarcity of unadulterated food, rising issue of global warming is _____. a. Stop using these resources. b. Sustainability c. to make people aware d. None of the above	1
Q2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i	Assertion (A): Naturalist Intelligence involves the ability to identify and categories living things and understands other features of the natural world. Reasoning (B): This type of intelligence is exhibited by poets, novelists, and journalists. (a) Both A and R are correct and R is the correct explanation of A (b) Both A and R are correct but R is NOT the correct explanation of A (c) A is correct but R is not correct (d) A is not correct but R is correct	1
ii	Results based on discriminatory assumptions made while developing algorithms by the developer are called as _____. a. AI Ethics b. AI Bias c. Test Data d. Training Data	1
iii	Statement A: Neural networks are made up of layers of neurons. Statement B: Human brain consists of millions of neurons. a. Only Statement A is correct b. Only Statement B is correct c. Both the statements are correct d. None of the statements is correct	1
iv	Neural Network is a mesh of multiple _____ layer.	1
v	Which of the following is a valid file extension for Notepad file? a. .jpg b. .doc c. .text d. .txt	1

vi	are the words which occur very frequently in the corpus but do not add any value to it. a. Special Characters b. Stopwords c. Roman Numbers d. Useless Words	1
Q3 Answer any 5 out of the given 6 questions on (1 x 5 = 5 marks)		
i	Which of the following is Not AI? i. Air Conditioner ii. Automatic Washing Machine iii. Self-Driving Car iv. Smart TV a. i, ii & iii b. i & ii c. i, ii & iv d. iii & iv	1
ii	Data science is the processing of diverse sets of data through? a. Organizing data b. Processing data c. Analyzing data d. All of the above	1
iii	Which is one of the significant data science skills? a. Statistics b. Data Visualization c. Machine Learning d. All of the above	1
iv	Which of the following tasks involves identifying and locating objects within an image? a. Image compression b. Feature extraction c. Object detection d. Image enhancement	1
v	Identify the type of Chabot with the information given below: These bots work on pre-programmed instructions inside the application/machine and are generally easy to develop. They are deployed in the customer care section of various companies. Their job is to answer some basic queries that they are coded for and connect them to human executives once they are unable to handle the conversation.	1
vi.	The percentage of true positive cases versus all the cases where the prediction is true is defined as a. Precision b. Accuracy c. F1 Score d. None of these	1

Q4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i	Aaron is facing issues in his internet connectivity at home. He called customer care service; A virtual assistant took his complaint and replied with an acknowledgement number. What is the AI technology used by Internet service providers? a. Data Science b. Machine Learning c. Computer Vision d. Natural Language Processing	1
ii	Statement 1: F1 score is evaluated based on precision or recall. Statement 2: When the F1 score is 0, the model accuracy is 100% a. Both statement1 and statement 2 are correct. b. Both statement1 and statement 2 are incorrect. c. Statement 1 is correct but statement 2 is incorrect. d. Statement 1 is incorrect but statement 2 is correct.	1
iii	Accuracy, Recall, Precision, F1 Score are the parameters to calculate the efficiency under _____ . a) Data Acquisition b) Data Evaluation c) Data Testing d) Data Modelling	1
iv	A grayscale image represents intensity values ranging from 0 to _____ .	1
v	What is the process of extracting emotions within a text data using NLP called? a. Sentiment Analysis b. Emotional Data Science c. Emotional Processing d. Emotional Classification	1
vi	An AI system is provided information about the residences, such as their prices, square footage, and number of rooms, features and whether or not they have gardens, using the associated labels. We can now train the model to forecast the price of a new house using data from thousands of houses, their attributes and pricing. Name the illustration.	1
Q5	Answer any 5 out of the given 6 questions on Employability Skills (1 x 5 = 5 marks)	
i	A well-known news outlet distributes information on a variety of subjects, such as politics, the federal government, world affairs, sports and the most recent developments in film and television. Gathering, analyzing and disseminating information about current events is the responsibility of reporters. Additionally, the news channel allows viewers to submit breaking news, which the channel's correspondents will verify. However, it could be challenging to tell the difference between true and false news, which could confuse readers and endanger the reputation of a news organization. In order for the audience to have a high level of trust in the news channel, which of the following domains is employed to identify fake news? a. Computer Vision b. Matrices c. NLP d. Data Science	1

ii	Which evaluation parameter takes all the percentage of correct predictions out of all the observations?	1
iii	Which of the following is a primary color in the RGB color model? a. Yellow b. Cyan c. Green d. Magenta	1
iv	In text normalization, text from multiple documents and the term used for the whole textual data from all the documents altogether is known as _____. a. Corpus b. Tokens c. Lemma d. Stem	1
v	Order shipped! Expected delivery in three days! No of tokens in the above sentence is a. 10 b. 9 c. 7 d. 8	1
vi	An AI model has been developed to detect credit card fraud detections. The aim is not to miss any fraud transactions. Therefore, we want False-Negative to be as low as possible. In these situations, we can compromise with the low precision, but _____ should be high. a. Accuracy b. F1 score c. Recall d. Confusion matrix	1

SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):

Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks)

Answer each question in 20 – 30 words

Q.6	Feedback is the final component and one of the most important factors in the process of communication. Give two reasons to justify why feedback is important.	2
Q.7	What is the importance of setting goals in life?	2
Q.8	Aisha is searching some contents related to sports in a search engine but during searching help her about how to protect her data while using the internet?	2
Q.9	List two qualities of successful Entrepreneurs.	2
Q.10	How can we achieve Sustainable development?	2

Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)

Q.11	“In the coming years, AI-enabled machines will replace all the people who work as laborers.” Justify this statement.	2
Q.12	What is targeted advertising? Explain with an example	2

Q.13	Suhana works for a company wherein she was assigned the task of developing a project using AI project cycle. She knew that the first stage was scoping the problem. Help her list the remaining stages that she must go through to develop the project.	2
Q.14	Explain the difference between grayscale and RGB images.	2
Q.15	Write the sentence segmentation for the following text: “Raj and Vijay are best friends. They play together with other friends. Raj likes to play football but Vijay prefers to play online games. Raj wants to be a footballer. Vijay wants to become an online gamer”	2
Q.16	Draw the confusion matrix for the following data • the number of true positive = 150 • the number of true negative = 67 • the number of false positive = 72 • the number of false negative = 190	2

Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)

Q.17	After class 12 Rahul wanted to join for AI course. His parents didn't know much about its domains. Explain to them the domains of AI.	4													
Q.18	Draw the 4Ws problem canvas and explain each one of them briefly.	4													
Q.19	Differentiate between rule-based and learning-based AI modelling approaches.	4													
Q.20	Consider the text of following documents: Document 1 : Sahil likes to play cricket Document 2 : Sajal likes cricket too Document 3: Sajal also likes to play basketball Apply all four steps of the bag of words of NLP on the above given documents and generate output.	4													
Q.21	A social media company has developed an AI model to predict which users are likely to churn (cancel their account). During testing, the AI model came up with the following predictions. <table border="1"><tr><th colspan="2" rowspan="2">Confusion Matrix</th><th colspan="2">Reality</th></tr><tr><th>yes</th><th>no</th></tr><tr><th rowspan="2">Predicted</th><th>yes</th><td>60</td><td>25</td></tr><tr><th>no</th><td>5</td><td>10</td></tr></table> (i) Calculate precision, recall and F1 Score for churn prediction (ii) How many total tests were performed in the above scenario?	Confusion Matrix		Reality		yes	no	Predicted	yes	60	25	no	5	10	4
Confusion Matrix				Reality											
		yes	no												
Predicted	yes	60	25												
	no	5	10												

